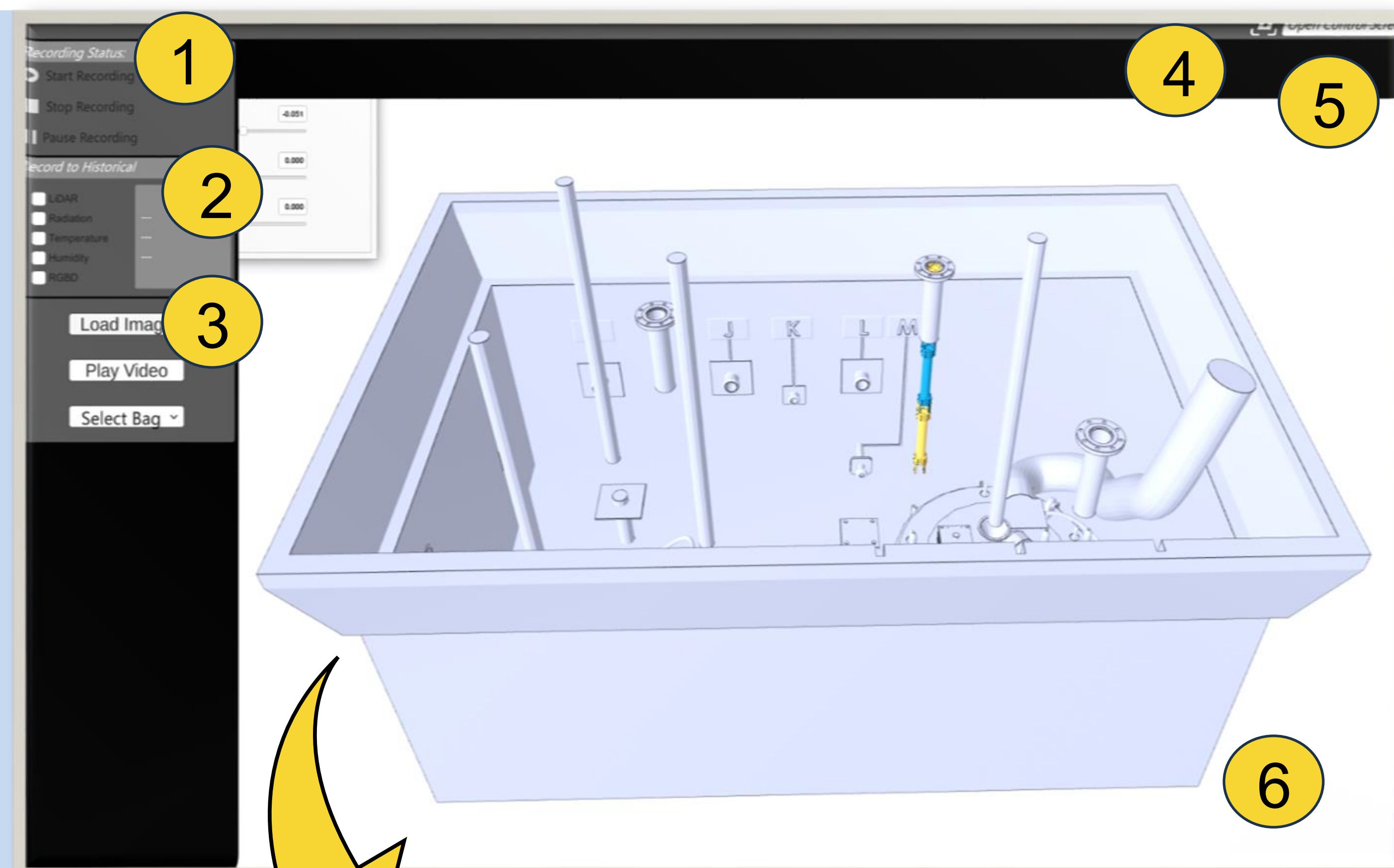


Problem

Nuclear waste pits require regular inspection to verify their integrity and support necessary cleanup. Current inspections are largely manual, exposing technicians to potentially radiated environments and costing over \$500,000 per pit inspection.

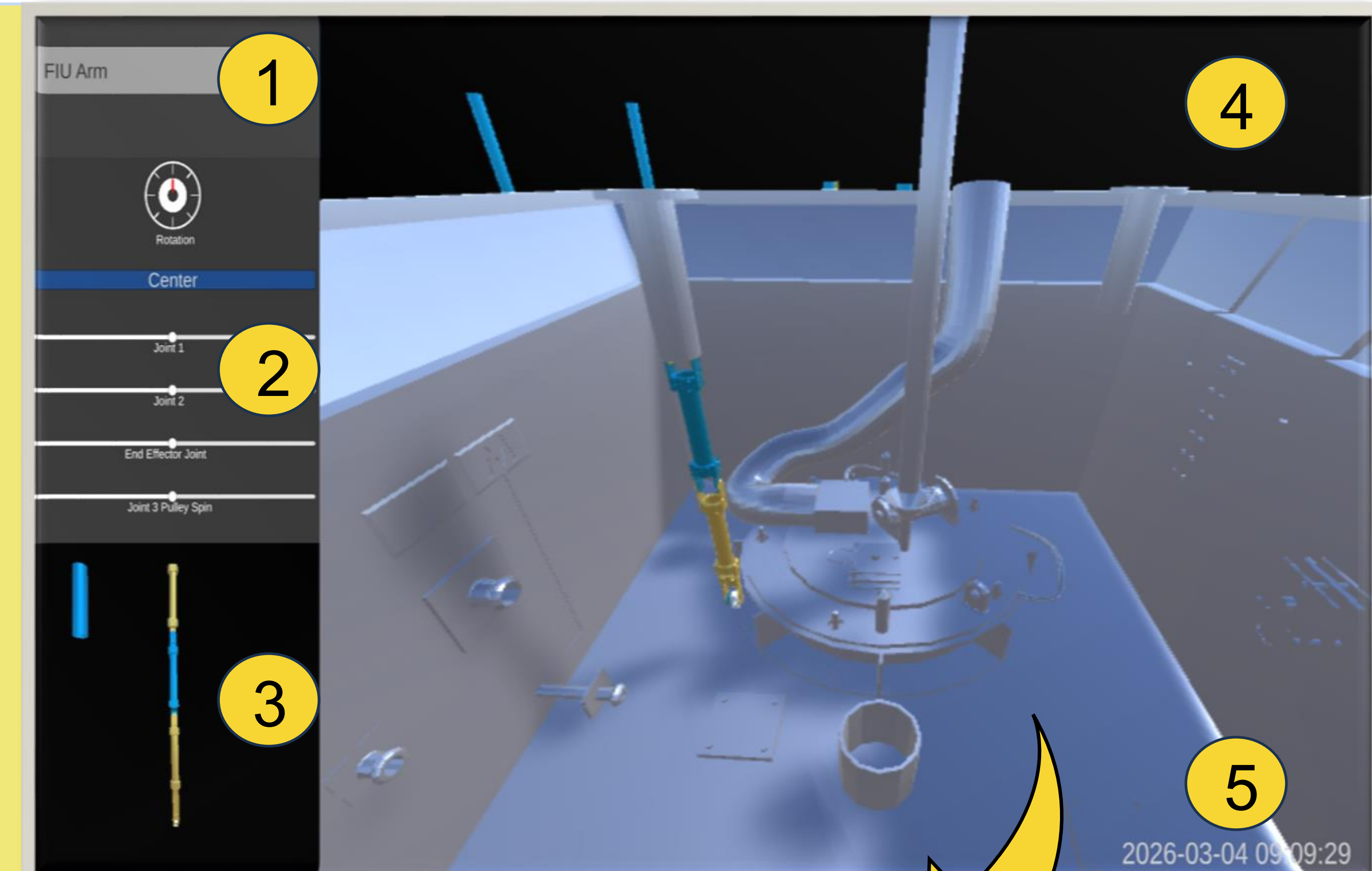
Solution

INL is developing a robotic platform for pit inspections, reducing operator exposure while enabling in-pit data collection. Florida International University developed the robotic arm while BYU developed the Unity-based interface for control and visualization.



Design Requirements

- Real-time visualization: seamless switch between camera and LiDAR point cloud with low-latency
- Accurate sensor overlays: independently toggleable RGB-D & LiDAR
- Operator control: situational awareness with one-screen access to start and stop the robot.
- Review and Export: gallery and replay with export photos/videos



Visualization Screen

- 1 – Start, Pause, Stop recording
- 2 – Historical Data Toggles of LiDAR, Radiation, Temp, Humidity & RGBD
- 3 – Historical Data Loading
- 4 – Take capture of visualization screen
- 5 – Open Control Screen
- 6 – Visualization Panel for robot's POV

Conclusion

A robotic inspection system can improve the safety, speed, and repeatability of nuclear waste pit inspections while reducing the need for personnel to enter radiation-controlled areas. The Unity based interface is a key enabler: it lowers operator workload, supports consistent execution of core inspection tasks, and brings critical inspection data into a single, clear view for real-time decisions and post-mission review. Together, the platform and UI increase overall inspection effectiveness while helping reduce cost and operational risk.

Control Screen

- 1 – Switch between robots
- 2 – Control robot joints
- 3 – View of robot position based on commands
- 4 – Visualization of robot within its environment
- 5 – Timestamp

